

Question 1

We are given:

$$r(\theta) = 12\cos(3\theta), r \geq 0.$$

(a)

A single lobe is traced where:

$$r \geq 0 \Rightarrow \cos(3\theta) \geq 0.$$

Thus,

$$-\frac{\pi}{2} \leq 3\theta \leq \frac{\pi}{2} \Rightarrow -\frac{\pi}{6} \leq \theta \leq \frac{\pi}{6}.$$

(b)

In polar coordinates:

$$dA = r \, dr \, d\theta.$$

Thus, the area is:

$$A = \int_{-\pi/6}^{\pi/6} \int_0^{12\cos(3\theta)} r \, dr \, d\theta.$$

(c)

Step 1: Inner integral

$$\int_0^{12\cos(3\theta)} r \, dr = \frac{1}{2} (12\cos(3\theta))^2 = 72\cos^2(3\theta).$$

Step 2: Outer integral

$$A = \int_{-\pi/6}^{\pi/6} 72 \cos^2(3\theta) \, d\theta.$$

Use identity:

$$\begin{aligned} \cos^2 x &= \frac{1 + \cos(2x)}{2}. \\ A &= 72 \int_{-\pi/6}^{\pi/6} \frac{1 + \cos(6\theta)}{2} \, d\theta = 36 \int_{-\pi/6}^{\pi/6} (1 + \cos(6\theta)) \, d\theta. \end{aligned}$$

Step 3: Compute

$$A = 36 \left[\int_{-\pi/6}^{\pi/6} 1 \, d\theta + \int_{-\pi/6}^{\pi/6} \cos(6\theta) \, d\theta \right].$$

$$A = 36 \cdot \pi/3 = 12\pi$$

Question 2

Sol (a) Minimize $f(x, y) = x^2 + 4y^2 - 4x + 8y$
subject to $x + 2y = 3$

$$L(x, y, \lambda) = x^2 + 4y^2 - 4x + 8y - \lambda(x + 2y - 3)$$

$$\frac{\partial L}{\partial x} = 2x - 4 - \lambda$$

$$\frac{\partial L}{\partial y} = 8y + 8 - 2\lambda$$

$$\frac{\partial L}{\partial \lambda} = -x - 2y + 3$$

$$\frac{\partial L}{\partial x} = 0$$

$$2x - \lambda = 4 \rightarrow (i)$$

$$\frac{\partial L}{\partial y} = 0$$

$$8y - 2\lambda = -8$$

$$4y - \lambda = -4 \rightarrow (ii)$$

$$\frac{\partial L}{\partial \lambda} = 0$$

$$x + 2y = 3 \rightarrow (iii)$$

$$\lambda = 2x - 4 \rightarrow (a)$$

$$\lambda = 4y + 4 \rightarrow (b)$$

If Lagrangian Ftn is +ive

$$L(x, y, \lambda) = x^2 + 4y^2 - 4x + 8y + \lambda(x + 2y - 3)$$

$$\frac{\partial L}{\partial x} = 2x - 4 + \lambda = 0 \rightarrow (i)$$

$$\frac{\partial L}{\partial y} = 8y + 8 + 2\lambda = 0 \rightarrow (ii)$$

$$\frac{\partial L}{\partial \lambda} = x + 2y - 3 = 0 \rightarrow (iii)$$

From eq (i)

$$\lambda = 4 - 2x \rightarrow (a)$$

From eq (ii)

$$4y + 4 + \lambda = 0$$

$$\lambda = -4y - 4 \rightarrow (b)$$

Equate (a) & (b)

$$4 - 2x = -4y - 4$$

$$2x = 8 + 4y$$

$$x = 4 + 2y \rightarrow (c)$$

Equate (a) & (b)

$$2x - 4 = 4y + 4$$

$$2x = 8 + 4y$$

$$x = 4 + 2y \rightarrow \textcircled{c}$$

Substitute (c) in (iii) we get

$$4 + 2y + 2y = 3$$

$$4 + 4y = 3$$

$$4y = -1$$

$$y = -\frac{1}{4}$$

Substitute y in (c)

$$x = 4 + 2\left(-\frac{1}{4}\right)$$

$$x = \frac{7}{2}$$

Substitute into constraint

$$(4 + 2y) + 2y = 3$$

$$4 + 4y = 3$$

$$y = -\frac{1}{4}$$

Substitute value of y in (c)

$$x = 4 + 2\left(-\frac{1}{4}\right) = \frac{7}{2}$$

$$\lambda = 4 - 2\left(\frac{7}{2}\right) = -3$$

$$(x^*, y^*) = \left(\frac{7}{2}, -\frac{1}{4}\right), \lambda = -3$$

$$f\left(\frac{7}{2}, -\frac{1}{4}\right) = -\frac{7}{2}$$

Hessian of f is +ive def. ftn
is strictly convex so

$\left(\frac{7}{2}, -\frac{1}{4}\right)$ is global min.

$$(b) \quad \lambda = 2\left(\frac{7}{2}\right) - 4 \Rightarrow \lambda = 3$$

$$f\left(\frac{7}{2}, -\frac{1}{4}\right) = -\frac{7}{2}$$

$$(c) \quad \text{Hessian of } f : \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 0 & 8 \end{bmatrix}$$

Hessian is positive definite so f is strictly convex. So $(x^*, y^*) = \left(\frac{7}{2}, -\frac{1}{4}\right)$ is unique global minimizer. $f^* = -\frac{7}{2}, \lambda = 3$.

Now applying Bordered Hessian Test to verify nature of C.P

$$g(x, y) = x + 2y - 3 = 0$$

$$g_x = 1, \quad g_y = 2$$

$$H_B = \begin{bmatrix} 0 & g_x & g_y \\ g_x & f_{xx} & f_{xy} \\ g_y & f_{yx} & f_{yy} \end{bmatrix}$$

$$H_B = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 8 \end{bmatrix}$$

$$|H_B| = \begin{vmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 8 \end{vmatrix} = -16$$

$|H_B| < 0$ so the C.P is L.M in

Q #3

$$P(x, y) = 4x - x^2 - y^2$$

$$x^2 + y^2 = 5 \quad (\text{constraint})$$

critical points:-

$$P_x = 4 - 2x = 0$$

$$P_y = -2y = 0$$

$$\boxed{x = 2}$$

$$\boxed{y = 0}$$

(2, 0) critical point

$$x^2 + y^2 = 5$$

$$4 + 0 \neq 5$$

inside the circle

not on the boundary. 8

Maximize on the boundary:-

$$x^2 + y^2 = 5$$

$$y^2 = 5 - x^2$$

$$P(x, y) = 4x - x^2 - (5 - x^2)$$

$$P(x, y) = 4x - \cancel{x^2} - 5 + \cancel{x^2}$$

$$P_x = 4$$

$$P_y = 0$$

[15 marks]

$$P(x) = 4x - 5$$

$$\boxed{x^2 < 5}$$

from constraint

$$-\sqrt{5} \leq x \leq \sqrt{5}$$

The function $4x - 5$
is linear and increasing
will give maximum
on $x = \sqrt{5}$ 6

$$y^2 = 5 - (\sqrt{5})^2 = 0$$

$$\boxed{y = 0}$$

$$(\sqrt{5}, 0)$$

$$P(\sqrt{5}, 0) = 4\sqrt{5} - 5$$

↓

maximum on
boundary. 01

Question 4

Sol

O.F : Distance b/w (x, y) & $(1, 2)$ squared is

$$f(x, y) = (x-1)^2 + (y-2)^2$$

Constraint : A circle of radius 1 centered at origin is

$$g(x, y) = x^2 + y^2 - 1 \leq 0$$

(a) KKT Conditions

* Lagrangian : $L(x, y, \lambda) = (x-1)^2 + (y-2)^2 + \lambda (x^2 + y^2 - 1)$

* Stationarity :

$$\frac{\partial L}{\partial x} = 2(x-1) + 2\lambda x = 0 \Rightarrow x(1+\lambda) = 1 \Rightarrow x = \frac{1}{1+\lambda} \rightarrow (i)$$

$$\frac{\partial L}{\partial y} = 2(y-2) + 2\lambda y = 0 \Rightarrow y(1+\lambda) = 2 \Rightarrow y = \frac{2}{1+\lambda} \rightarrow (ii)$$

* Complementary Slackness : $\lambda(x^2 + y^2 - 1) = 0$

* Primal Feasibility : $x^2 + y^2 - 1 \leq 0$ * Dual Feasibility : $\lambda \geq 0$

(b) Solving for x^*, y^* & Condition $\rightarrow$ complementary slackness requires that either constraint is active or multiplier is 0.

Case 1 Constraint is inactive ($\lambda = 0$)

If $\lambda = 0$ then $x = \frac{1}{1+0} = 1$

$$y = \frac{2}{1+0} = 2$$

check Primal Feasibility : $(1)^2 + (2)^2 - 1 \leq 0$
 $4 \not\leq 0$

Since $4 > 0$, the pt $(1, 2)$ is outside the feasible region
So inactive sol is invalid

Case 2 If constraint is active

$$x^2 + y^2 - 1 = 0$$

$$x^2 + y^2 = 1 \rightarrow (iii)$$

Substitute (i) & (ii) in (iii)

$$\frac{(1)^2}{(1+\lambda)^2} + \frac{(2)^2}{(1+\lambda)^2} = 1$$

$$\frac{5}{(1+\lambda)^2} = 1$$

$$(1+\lambda) = \sqrt{5}$$

$$\lambda = \sqrt{5} - 1$$

$$\lambda = 1.236$$

$$x^* = \frac{1}{\sqrt{5}} \approx 0.447$$

$$y^* = \frac{2}{\sqrt{5}} \approx 0.894$$

check Dual feasibility: Since $\lambda \approx 1.236 \geq 0$, this is valid sol.

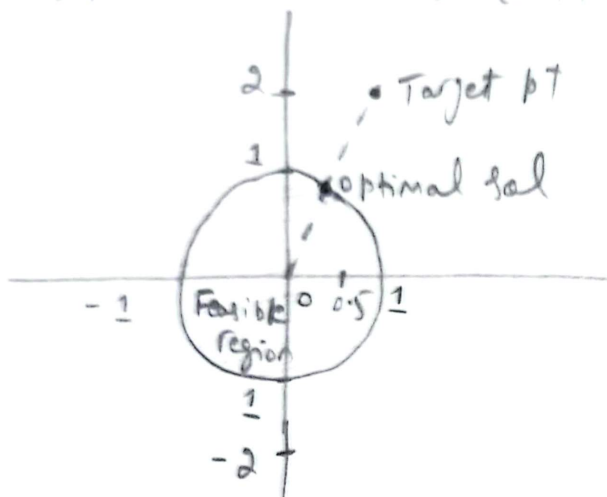
Substitute λ back into eqs for x & y

$$x^* = \frac{1}{\sqrt{5}} \approx 0.447$$

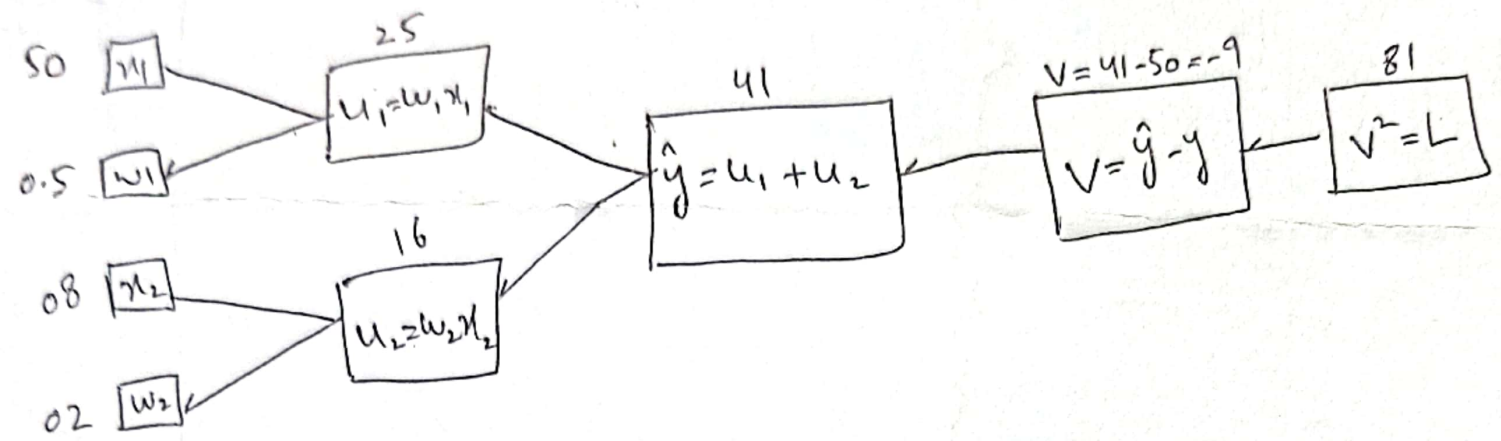
$$y^* = \frac{2}{\sqrt{5}} \approx 0.894$$

optimal sensor location is $(0.447, 0.894)$

(c)



Sol. Question 5



Forward Phase

$$L = 81$$

Backward Phase

$$\begin{aligned} \frac{\partial L}{\partial w_1} &= \frac{\partial L}{\partial V} \frac{\partial V}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial u_1} \frac{\partial u_1}{\partial w_1} \\ &= (2V)(1)(1)(x_1) = (2)(-9)(50) \\ &= -900 \end{aligned}$$

$$\begin{aligned} \frac{\partial L}{\partial w_2} &= \frac{\partial L}{\partial V} \frac{\partial V}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial u_2} \frac{\partial u_2}{\partial w_2} \\ &= (2V)(1)(1)(x_2) = (2)(-9)(8) = -144 \end{aligned}$$

⇒ CPU prediction increases if active request increases and CPU prediction is less sensitive to memory usage.

Q #6

[10 marks]

$$T(x, y) = x^2 + y^2 - 4x - 6y + 12$$

(x_0, y_0)

$$x_1 = (0, 0)$$

$$\nabla T(x, y) = \begin{bmatrix} 2x - 4 \\ 2y - 6 \end{bmatrix}, \quad \nabla T(0, 0) = \begin{bmatrix} -4 \\ -6 \end{bmatrix}$$

$$S_1 = -\nabla T(0, 0) = \begin{bmatrix} 4 \\ 6 \end{bmatrix} \rightarrow \text{search direction} \quad \underline{3}$$

$$H = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \rightarrow \text{Hessian} \quad \underline{3}$$

$$\lambda_1 = \frac{S_1^T S_1}{S_1^T H S_1} = \frac{\begin{bmatrix} 4 & 6 \end{bmatrix} \begin{bmatrix} 4 \\ 6 \end{bmatrix}}{\begin{bmatrix} 4 & 6 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 6 \end{bmatrix}} = \frac{52}{104}$$

$$\lambda_1 = 1/2 \rightarrow \text{learning rate} \quad \underline{3}$$

$$x_2 = x_1 + \lambda S_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 3 \end{bmatrix} \rightarrow \text{updated parameters} \quad \underline{01}$$